

Increasing market transparency of residential air-to-water heat pumps in Europe by clustering the appliances

Ugne Potthoff^a, Felix Morlock^a, Felix Wortmann^b

^aDs-MTEC, ETH Zürich, Rämistrasse 101, 8092 Zürich, Switzerland

^bITEM-HSG, Dufourstrasse 40a, 9000 St. Gallen, Switzerland

Abstract

The transition to low-carbon heating requires not only efficient technologies but also transparent markets that enable informed policy and consumer decisions. Yet, users and installers face challenges in comparing air-to-water heat pumps (AWHPs) due to complex technical parameters and inconsistent model naming across countries and online platforms. This study aims at helping to navigate in choosing the heat pump by improving the understanding of how heat pumps are priced in Europe. Using the open-source *hplib* library, a dataset of 696 Keymark-certified AWHPs from 13 manufacturers was compiled, covering units up to 13 kW using low-GWP refrigerants R32 and R290. Retail prices were collected through automated web scraping of European online vendors and matched to certified technical parameters. Price modelling shows that the main price drivers are rated power, storage size and refrigerant type. Together with noise emission, standby power, reference COP (coefficient of performance), and tank configuration, these factors explain most of the observed price variation ($R^2 = 0.84$). The three dominant drivers substantially outweigh the influence of COP, which contributes only marginally to price formation. Unsupervised clustering identified five distinct product groups differentiated mainly by brand origin (EU vs. non-EU) and configuration (monoblock vs. split). European heat pumps show higher maximum supply temperatures and lower noise levels, indicating premium and suitability for retrofits, while non-EU heat pumps focusing on affordability. The efficiency remains similar between regional groups (SCOP = 4.70 EU vs. 4.76 non-EU). However, European models are priced higher (€1,362 / kW vs. €1,023 / kW). Overall, this study integrates model matching, price modeling, and clustering to contribute to a transparent, data-driven European AWHP market. The findings enhance product comparability and provide evidence to support policymakers and consumers in accelerating the heat-pump transition.

© HPC2026.

Selection and/or peer-review under the responsibility of the organizers of the 15th IEA Heat Pump Conference 2026.

Keywords: Air-to-water heat pumps; Price drivers; Product naming; SCOP; Market transparency; European heating market; Web-scraping;

1. Introduction

Heat pumps (HPs) have become central to the decarbonization of Europe's building stock, providing an efficient alternative to fossil-fuel-based heating systems. Their market penetration has grown rapidly, with more than 24 million units now installed alone in Europe [1]. Air-to-water heat pumps (AWHPs) in particular play a crucial role in residential heating transitions due to their lower installation costs compared to ground- or water-source systems. Despite policy support and improving technology, adoption remains uneven, and market transparency continues to be a barrier to wider deployment [2].

One of the central challenges in the European air-to-water heat pump (AWHP) market is the large variation in retail prices among technically comparable products. A survey by a German consumer advice center analyzing 160 installation offers revealed price differences of up to a factor of three for similar systems, creating significant uncertainty and distrust among potential adopters [3]. Such discrepancies hinder consumer confidence and slow down the market uptake of low-carbon heating technologies.

A key reason for these inconsistencies lies in the limited market transparency. Due to the technical complexity of heating systems and the dominance of B2B-oriented sales channels, homeowners rarely receive standardized product information. Moreover, non-uniform product naming across countries, sales platforms, and even within manufacturer catalogues prevents direct price comparison. Two heat pumps with identical certified performance parameters may appear under different commercial names or configurations, obscuring comparability for consumers, installers, and analysts alike. This fragmentation fosters information asymmetry, where buyers cannot distinguish whether price differences reflect real performance or simply branding and configuration choices.

Beyond these data inconsistencies, there remains limited empirical understanding of what actually drives heat pump prices. Manufacturers often attribute higher prices to superior efficiency, commonly expressed by the Seasonal Coefficient of Performance (SCOP). However, empirical evidence supporting this relationship is scarce. Previous studies have primarily examined cost structures or technical performance in isolation. For example, Ban and Bungău [4] and Paiho et al. [5] identified strong cost differences between system types but did not link them to retail pricing. Bianco et al. [6] focused on primary energy and life-cycle efficiency, while Miara et al. [7] emphasized discrepancies between laboratory and field performance. Yet, the interaction between efficiency indicators, configuration type, and retail prices across manufacturers remains insufficiently quantified. Understanding this relationship is essential for designing transparent, data-driven policies and for enabling fair comparison in the rapidly expanding European heat pump market.

While previous research has focused on efficiency metrics such as COP and SCOP or life-cycle economics, little attention has been given to data-driven market analysis linking certified product data with real-world retail prices. This study addresses that gap by combining Keymark certification data with retail prices collected through automated web scraping of European online markets. To overcome naming inconsistencies, a reference dictionary was developed to align identical products listed under different identifiers, creating an unified dataset for analysis.

Previous research on heat pumps has focused mainly on efficiency metrics such as COP and SCOP or on life-cycle economics, while little work has linked certified technical data to real-world retail prices. This study addresses that gap by combining HP Keymark data [8] with retail prices obtained through automated web scraping of European online markets. Because identical models often appear under different commercial identifiers, a reference dictionary was developed to harmonize naming and create a unified dataset. Using this consolidated data, the study aims to quantify retail price variation across certified AWHPs, identify the technical and market-related factors that explain price differences, and detect structural product clusters based on combined technical and commercial attributes. This integrated approach provides new empirical insight into how performance characteristics, configuration types, and market practices shape AHP pricing.

2. Methods

2.1. Data and scope

We constructed a dataset of residential air-to-water heat pumps (AWHPs) from the Heat Pump Library (hplib) based on Keymark certificates (snapshot: February 2025) [8,9]. Certificates provide harmonized technical specifications, including model identifiers, SCOP, rated heating capacity at low temperature, outdoor sound power level, and maximum water temperature. The study focuses on AWHPs with nominal capacity ≤ 13 kW and using R32 or R290 refrigerants to represent the European residential segment [10,11]. Duplicate certificate entries and non-AHP technologies were removed. The resulting technical base covers models from 13 manufacturers.

2.2. Retail price collection

Market prices were collected from European e-commerce platforms via automated web scraping. For each listing we extracted manufacturer, model string, price, currency, URL, and scrape date. Prices were standardized to EUR, exact duplicates were dropped (keeping the most recent observation), and entries less than €1,000 were excluded as likely accessories or non-comparable items.

2.3. Model configuration decoding

To enable like-for-like comparisons, we systematically annotated each Keymark-certified model with configuration attributes that affect price: monoblock vs. split, presence of a hydraulic module, and integrated

domestic hot-water tank (recording tank capacity where available). Classification combined certificate metadata with manufacturer documentation and major reseller listings; when encoded hints were present (e.g., suffixes indicating “m/s” or tank size), they were resolved and verified against source pages. The resulting configuration labels were merged with the technical records, allowing products to be grouped not only by performance (e.g., power) but also by system architecture - an essential prerequisite for unbiased price comparisons and subsequent segmentation.

2.4. Cross-source model matching

Market listings frequently use variant model strings that differ from Keymark identifiers (regional codes, formatting, added descriptors). We therefore built a reference dictionary to reconcile vendor/model strings to their corresponding Keymark entries, as shown in Table 1. Matching applied token normalization (lower-casing, removal of special characters and regional suffixes) and rule-based reconciliation; ambiguous or low-confidence pairs were manually reviewed. This iterative procedure - refining rules whenever unmatched items were detected - produced a unified dataset that reliably links certified specifications with observed retail prices, forming the basis for further data analysis.

Table 1: Exemplary name matches between scraped vendor data and official hplib entries. For each manufacturer, a selected row shows how the scraped Model/Type column was matched with the hplib Titel column.

Manufacturer	Model (Keymark database)	Title (Keymark database)	Model (Scraped data)
Bosch	CS5800i/6800iAW 10/12 OR	CS5800iAW 10 ORM-T	BOSCH Luftwärmepumpen-Paket BOPA CS788 AW 10 OR-T, CS5800iAW 12 M, 7739623429
Daikin	ALTHERMA 4 H ECH2O 08–10 kW 500L (3ph)	EPSK08AW1 EPSX(B)10P50A	Daikin Altherma 4 H ECH2O Set EPSK-08AW1 + EPSX-10P50A Hydrobox/Speicher 500 L WiFi R290 Heizstab 9 kW
Johnson Hitachi	Yutaki M 5.0 HP R32 (mono)	RASM-5VR1E – with cooling kit	Hitachi Yutaki M RASM-5VR1E Monoblock Wärmepumpe zum Heizen / Kühlen / Brauchwasserbereitung 12,0 kW
...			
... full table available in the supplementary GitHub repository.			

2.5. Binarized region

To account for potential regional patterns, the origin of each manufacturer was binarized into two categories European (EU) and non-European (non-EU) - based on the brand’s headquarters location. This simplification ensured a neutral representation of manufacturer origin.

2.6. Price modelling

To quantify how technical attributes influence heat pump prices, several regression and machine-learning models were evaluated using five-fold cross-validation on an 80/20 train-test split. The feature set included a broad range of technical and categorical variables: rated heating capacity at low and medium temperatures, refrigerant type and charge, indoor and outdoor sound power levels, maximum supply temperature, standby power, SCOP, electrical heater capacity, COP at a reference point (-7 °C / 52 °C), configuration type (monoblock or split), storage tank size and configuration, and brand-origin region. Linear models (OLS, Ridge) and ensemble algorithms (Random Forest and Gradient Boosting) were tested. Model performance was compared using out-of-sample R², RMSE, and MAE, and feature subsets of varying dimensionality (top 3, 5, and 7 predictors) to assess model robustness. Based on these evaluations, Gradient Boosting with grid-search hyperparameter optimization and cross-validation was selected as the final modelling approach. Feature importance was subsequently derived using impurity-based metrics and SHAP values to ensure transparent interpretation of how individual attributes contribute to price formation.

2.7. Market segmentation (clustering)

To uncover structural patterns in the European air-to-water heat pump market, we applied Gaussian Mixture Model (GMM) clustering on nine standardized criteria that jointly describe technical and commercial product characteristics, as shown in Table 2.

Table 2: Feature description used for clustering.

Feature	Description
SCOP	Seasonal Coefficient of Performance
Quietness	Inverse of outdoor noise level (dBA)
Value	Inverse of price per kW
Max temp	Maximum water heating temperature (°C)
Power	Rated power at low temperature (kW)
Affordability	Inverse of total price
Storage size	Size of installed water storage (L)
Region	Binarized origin of the manufacturer brand
Config	Monoblock / Split

The optimal number of clusters was determined by minimizing the Bayesian Information Criterion (BIC), with the Akaike Information Criterion (AIC) serving as a consistency check. Each product was assigned to one of the five Gaussian mixture clusters, and summary statistics (mean, median, and percentile values) were computed for key technical and price-related variables. To derive interpretable market archetypes, a rule-based labeling system was applied using percentile thresholds across multiple attributes, summarized in Table 3. Clusters were named according to their dominant characteristics - meaning that if a feature showed systematically similar values within a cluster, it served as the defining label, as summarized in the table below. Up to three dominant traits were combined to form concise and interpretable market labels, such as EU-Monoblock-Retrofit or non-EU-Split.

Table 3: Trait assignment criteria based on percentile thresholds. Traits are used to label each cluster with interpretable market personas.

Trait	Selection Criterion
Retrofit	Cluster average max. water temperature > 65 °C (suitable for older radiator systems)
Quiet	Average noise level below 20th percentile (low acoustic output)
EU	>80% of cluster items have manufacturers with EU origin
Non-EU	>80% of cluster items have manufacturers with non-EU origin
Monoblock	>80% of cluster items are monoblock heat pumps
Split	>80% of cluster items have manufacturers with EU origin

2.8. Reproducibility

All preprocessing, matching, and analysis scripts are reproducible and available in a public repository. The matched analysis dataset (Keymark specs + prices + configuration attributes) and code to regenerate figures and tables accompany this paper.

3. Results

3.1. Dataset overview

The final dataset comprised 696 certified air-to-water heat pumps (AWHPs) from 13 manufacturers, combining technical specifications from the Keymark certification database with retail prices collected via automated web scraping. Valid retail prices were successfully matched for 26.5% of all certified models, as shown in Table 4. Unmatched cases primarily resulted from: (1) inconsistent or unidentifiable product names and technical specifications, (2) incomplete system configurations where models are sold as separate components and a comparable bundle couldn't be created, and (3) units that are not yet available or no longer on the market.

Table 4: Overview of selected HPs matched with a price. The table shows the number of HPs from the filtered hplib database, the number of scraped items associated with the corresponding manufacturer, the number of matches, and the matched percentage.

Manufacturer	# Database	# Scraped	# Matched	% Matched
Alpha Innotec	27	257	22	85.19 %
Bosch	83	732	57	68.67 %
Buderus	84	1128	72	86.71 %
Daikin	875	1025	102	11.66 %
Johnson Hitachi	139	49	22	15.83 %
LG	80	195	58	72.50 %
Mitsubishi	610	480	123	20.16 %
Panasonic	137	215	48	35.04 %
Samsung	148	90	38	25.68 %
Toshiba	162	39	22	13.58 %
Vaillant	89	635	38	42.70 %
Viessmann	171	1015	71	41.52 %
WOLF	23	657	23	100.00 %
Total	2628	6517	696	26.48 %

Across all models, the average retail price was EUR 8,898 (range: EUR 2,725–17,499). The mean SCOP was 4.72 (3.84–5.45), and the average rated power was 8.25 kW. Outdoor noise levels averaged 54.6 dBA, and the mean maximum water temperature was 59.4 °C (Figure 1).

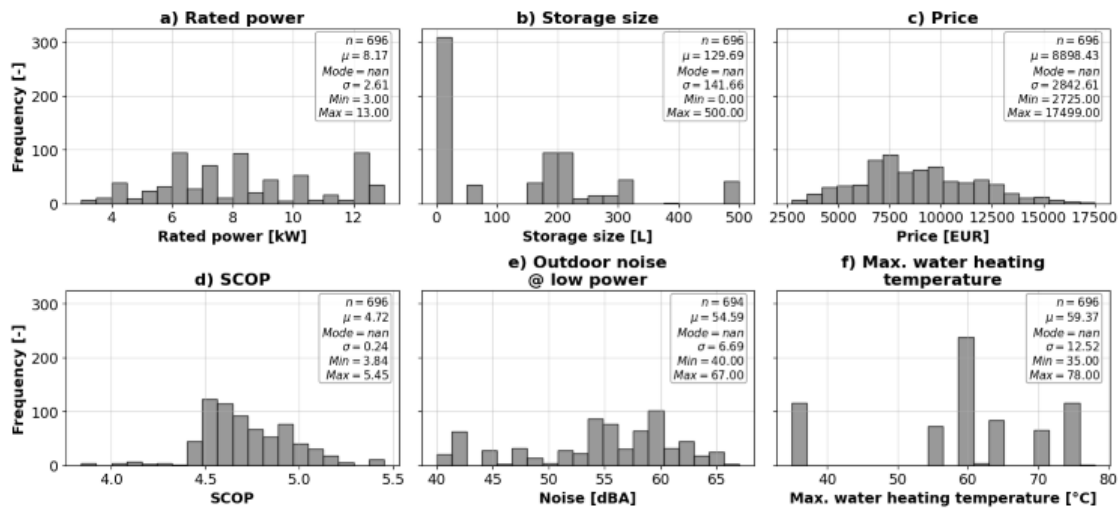


Figure 1: Descriptive statistics of key heat pump features of the scraped heat pump data ($n = 696$).

Two refrigerants dominated: R32 ($n = 446$) and R290 ($n = 250$). R290-based systems had higher prices (mean = EUR 10,746 vs. 7,863) and slightly higher SCOP (4.78 vs. 4.69). Mann-Whitney U tests confirmed significant differences for both price ($U = 87,526$, $p < 0.001$, $r = 0.47$) and SCOP ($U = 68,159$, $p < 0.001$, $r = 0.18$). However, refrigerant type is confounded with system architecture: all R290 models were monoblock systems, while R32 appeared in both split and monoblock designs.

3.2. Price modeling with Gradient Boosting model

The Gradient Boosting Regressor emerged as the best-performing model, achieving a strong predictive fit with an R^2 of 0.92 (train) and 0.83 (test), and mean absolute errors of €551 and €887, respectively. These results indicate high predictive accuracy and limited overfitting across the dataset.

The SHAP analysis revealed that rated power, storage size, and refrigerant type were the most influential determinants of retail price, while noise level, standby power, and COP at the reference point also contributed meaningfully (Figure 2). In contrast, SCOP and maximum water temperature showed only marginal effects. Rated power showed the strongest SHAP impact (€1,174.61), indicating that variation across the 3.44–13 kW range contributes more to predicted price differences than any other feature. Across all heat pumps in the dataset, the variation in storage-tank size (from 0 to 500 L) leads, on average, to about €960 difference in predicted price. Larger storage volumes, systems with R290 refrigerants and quieter designs were likewise associated with higher prices. The impact of each features on the price is shown in Table 5.

Table 5: Top 7 features ranked by mean absolute SHAP value, representing average impact on predicted price in EUR.

Feature	Mean SHAP Value (EUR)	Value ranges in the dataset
Rated power	1174.61	3.44 - 13.0 kW
Storage size	959.80	0 - 500 Liters
Refrigerant type	686.09	R32: 0, R290: 1
Noise	403.77	40 - 69 dBA
Standby power	398.19	0 - 60 W
COP at reference point	246.29	1.65 – 2.65
Tank configuration	206.89	Hydraulic module:0, with warm water tank: 1

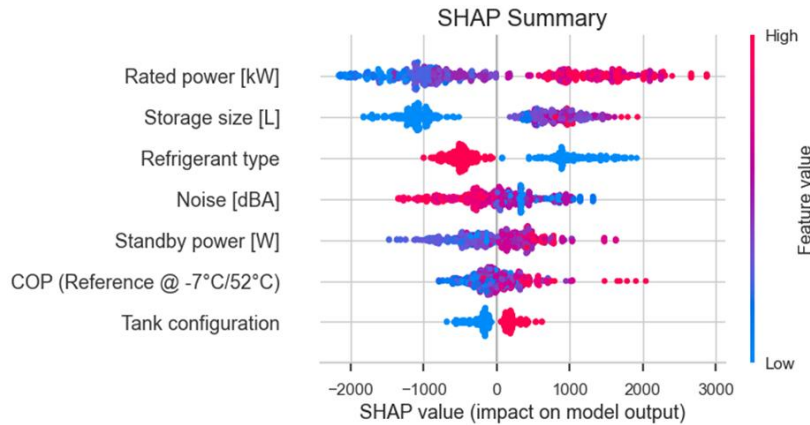


Figure 2: SHAP summary plot showing both global and local feature effects from the gradient boosting model. Each dot represents an individual sample; color indicates feature value, and position indicates the SHAP impact on predicted price.

3.3. Market segmentation

Unsupervised Gaussian Mixture clustering applied to standardized heat pump features (SCOP, noise level, price per kW, rated power, and maximum water temperature) revealed five distinct product groups within the dataset of 696 certified air-to-water heat pumps. Cluster validation using the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) confirmed the selection of five clusters as the best trade-off between compactness and separation (Figure 3).

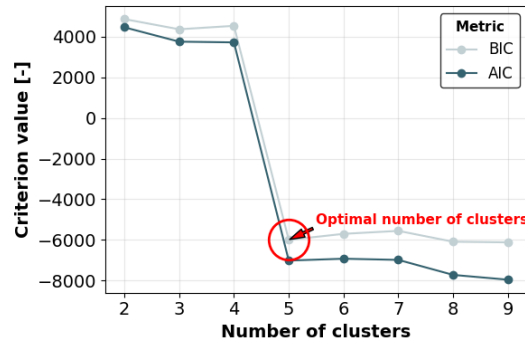


Figure 3: Optimal number of clusters for Gaussian Mixture clustering based on BIC and AIC. Both metrics reach their lowest values at $k = 5$, indicating that five clusters provide the best balance between model fit and parsimony.

The resulting clusters are primarily structured by manufacturer origin (EU vs. non-EU) and system configuration (monoblock vs. split). Despite no prior labelling, the algorithm naturally grouped European and non-European manufacturers, highlighting systematic differences in product design and market positioning. The radar plots in Figure 4 further illustrate these distinctions, showing that non-EU clusters emphasize affordability and standardization, while EU clusters focus on retrofit capability, quiet operation, and higher supply temperatures - reflecting divergent value propositions within the European heat pump market.

The non - EU clusters (Monoblock and Split, jointly 59.3 % of models) emphasize affordability and standardized configurations, with average prices below €9,000 and typical rated powers around 8-9 kW. These systems predominantly use R32 refrigerant and target the mass residential segment. In contrast, EU clusters (40.7 %) exhibit greater design diversity and specialization. The EU-Monoblock-Retrofit and Retrofit-Quiet groups feature higher maximum supply temperatures (66-71 °C) and lower noise levels (down to 41 dBA), catering to retrofit and comfort-oriented markets. The EU-Split cluster balances moderate efficiency (SCOP ≈ 4.8) with compact form factors and mid-range prices. The cluster features are summarized in Table 6.

Overall, the unsupervised clustering demonstrates that regional manufacturing origin and system configuration dominate market segmentation, while efficiency (SCOP) varies only modestly across clusters. The resulting archetypes - from non-EU-Split budget systems to EU-Monoblock-Retrofit-Quiet premium models - reflect distinct value propositions and consumer priorities within the European heat pump market.

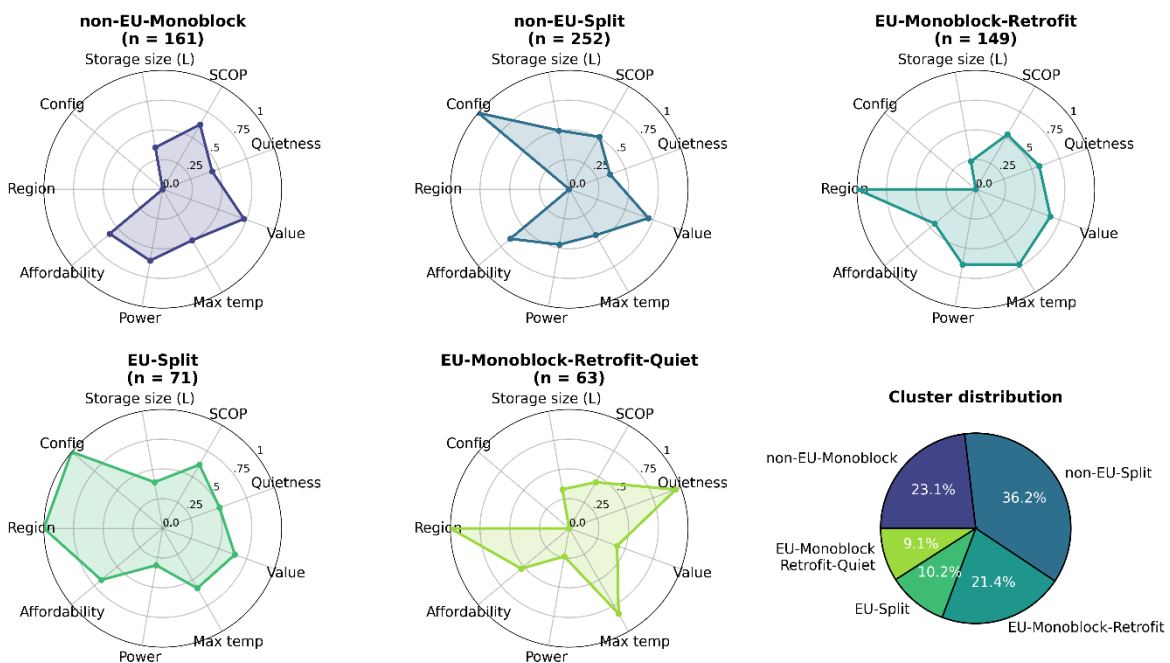


Figure 4: Radar plots illustrating the normalized feature profiles of each cluster (market segment), along with a pie chart showing the market share of these clusters.

Table 6: Detailed segment-level cluster profiles, including number of products, performance indicators, brand, and configuration traits for each cluster.

Attribute	non-EU-Monoblock	non-EU-Split	EU-Monoblock-Retrofit	EU-Split	EU-Monoblock-Retrofit-Quiet
Number of products	161 (23.1 %)	252 (36.2 %)	149 (21.4 %)	71 (10.2 %)	63 (9.1 %)
Average price [€]	8,894 ± 3,466	7,930 ± 1,710	10,851 ± 3,159	7,529 ± 2,117	9,707 ± 1,752
Price per kW [€/kW]	1,002 ± 410	1,046 ± 270	1,147 ± 287	1,202 ± 323	1,738 ± 418
SCOP	4.85 ± 0.28	4.66 ± 0.17	4.70 ± 0.28	4.84 ± 0.17	4.56 ± 0.01
Noise level [dBA]	56.2 ± 5.0	58.4 ± 3.5	52.5 ± 6.4	54.2 ± 5.5	41.3 ± 1.0
Rated power [kW]	9.3 ± 2.5	8.0 ± 2.3	9.6 ± 2.3	6.5 ± 1.6	5.7 ± 1.0
Max. temperature [°C]	56 ± 16	54 ± 11	66 ± 9	60 ± 0	71 ± 7
Refrigerants	R32 (94), R290 (67)	R32 (252)	R290 (120), R32 (29)	R32 (71)	R290 (63)

4. Discussion

This study provides the first large-scale, data-driven analysis linking certified technical specifications of AWHPs with real market prices across Europe. By combining Keymark certification data with retail prices obtained through automated web scraping, the research quantifies how product characteristics - rather than nominal efficiency ratings - shape consumer-facing costs. Results from a fine-tuned gradient boosting model ($R^2 = 0.87$, MAE app. €740) show that rated heating power, configuration, and noise level are the dominant price determinants, while SCOP contributes only marginally. Each additional kilowatt of heating capacity increases price by approximately €535-1,150, and monoblock systems with integrated tanks are on average €2,300 more expensive than comparable configurations.

Unsupervised clustering further revealed five distinct market segments primarily differentiated by manufacturer origin (EU vs. non-EU) and configuration type (monoblock vs. split). European clusters emphasize retrofit suitability, higher maximum water temperatures, and lower noise levels, whereas non-EU clusters focus on affordability and standard residential applications. Together, these findings highlight systematic design and pricing differences across global suppliers and contribute to improving market transparency for policymakers and consumers.

4.1. Implications

The analysis provides evidence on how technical and design-related factors influence heat pump prices in Europe. By focusing on the equipment cost component, the results offer insights directly relevant to policy instruments such as purchase subsidies or efficiency standards. The weak association between SCOP and price indicates that incentives based solely on laboratory efficiency metrics may not reflect real market dynamics. Instead, price formation appears to depend more strongly on rated power, configuration, and acoustic performance, suggesting that a broader set of evaluation criteria could improve the effectiveness of support schemes, particularly for retrofit applications.

The observed variation in price across manufacturer origin regions and configurations also suggests limited comparability in product information. Broader access to certified data, supported by consistent product identifiers and naming conventions, could improve the reliability of market analyses and consumer comparisons. Methodologically, the study shows that linking certification databases with retail data through open-source tools enables scalable and transparent market monitoring. This approach can be extended to other heating and appliance markets, providing a reproducible framework for evaluating product-level cost and performance drivers.

4.2. Limitations and future research

While this study provides statistically robust insights into price formation in the European air-to-water heat pump market, several limitations suggest directions for future work. Expanding retailer coverage to Southern and Eastern Europe would improve geographic representativeness and allow regional price comparisons. Incorporating temporal price data could further reveal seasonal or policy-driven dynamics. Future research may also benefit from text-mined product specifications or manufacturer documentation to capture attributes such as control sophistication, build quality, or design, which are not fully represented in certification data.

Data limitations remain a central challenge. Matching Keymark-certified models with retailer listings was constrained by non-standardized product naming, reducing the number of valid matches. Although this is unlikely to bias results directionally, it may limit representativeness toward models with clearer labeling. Furthermore, the analysis focuses exclusively on equipment prices, excluding installation, accessories, and operational costs. This controlled scope isolates product-level pricing behavior but should be considered when comparing with total system cost studies.

Finally, the absence of detailed metadata for advanced features (e.g., inverter type, defrosting mode, or control strategy) and the lack of real-world efficiency or installation data constrain explanatory power. Combining certified and market data with field measurements of installed costs and performance would allow more comprehensive techno-economic assessments and better inform policy design, payback analysis, and total cost of ownership modeling.

5. Conclusion

This study provides the first quantitative, data-driven assessment linking certified technical specifications of residential AWHPs with real market prices across Europe. By combining Keymark certification data with web-scraped retail prices, it offers an evidence-based view of how equipment attributes influence market valuation. The analysis demonstrates that rated power, configuration type, acoustic performance, and refrigerant choice are the dominant factors shaping product pricing, whereas SCOP - often used in marketing and policy metrics - plays only a minor role.

Unsupervised Gaussian mixture clustering revealed five distinct product segments, differentiated primarily by manufacturer origin (EU vs. non-EU) and system configuration (monoblock vs. split). European products tend to emphasize higher water outlet temperatures and lower noise, reflecting a focus on retrofit and comfort applications, while non-EU products are positioned toward affordability and standard residential use. Together, these results highlight structural differences in product design and pricing strategies among global suppliers.

Beyond empirical insights, the study contributes a reproducible open-data framework that integrates certified product databases with market data, helping to improve transparency and comparability in the European heat pump market. Future extensions combining equipment prices with installation, maintenance, and operational cost data could provide a more comprehensive picture of total system economics. Overall, the findings inform evidence-based policy design, support consumer-oriented transparency, and advance empirical understanding of how technology, performance, and value interact in the residential heat pump sector.

References

- [1] European Heat Pump Association (EHPA). EHPA Market Report 2024. Brussels: EHPA; 2024. Available from: https://www.ehpa.org/wp-content/uploads/2024/08/Executive-summary_EHPA-heat-pump-market-and-statistic-report-2024-2.pdf. Accessed 11 November 2025.
- [2] Toleikyte A, Roca Reina JC, Volt J, Carlsson J, Lyons L, Gasparella A, et al. The Heat Pump Wave: Opportunities and Challenges. Luxembourg: Publications Office of the European Union; 2023. <https://op.europa.eu/en/publication-detail/-/publication/94cf1914-10b1-11ee-b12e-01aa75ed71a1/language-en>. Accessed 11 November 2025.
- [3] Verbraucherzentrale Rheinland-Pfalz e.V. Luft-Wasser-Wärmepumpen: Eine Auswertung von 160 Angeboten aus Rheinland-Pfalz. Mainz: Verbraucherzentrale Rheinland-Pfalz; 2025. Available from: https://www.verbraucherzentrale-rlp.de/sites/default/files/2025-06/250605_vz-rlp_auswertung_wp_angebote.pdf. Accessed 11 November 2025.
- [4] Ban A, Bungău C. (2022). Estimation of the initial investment cost in systems equipped with heat pumps using regression analysis. Journal of Engineering Sciences and Innovation. 7. 293-304. https://jesi.astr.ro/wp-content/uploads/2023/03/2_Ban-Alexandra.pdf
- [5] Paiho S, Pulakka S, Knuuti A. Life-cycle cost analyses of heat pump concepts for Finnish new nearly zero-energy residential buildings. Energy and Buildings, Volume 150, 2017, Pages 396-402. <https://doi.org/10.1016/j.enbuild.2017.06.034>
- [6] Bianco V, Scarpa F, Tagliafico LA. Estimation of primary energy savings by using heat pumps for heating purposes in the residential sector. Applied Thermal Engineering, Volume 114, 2017, Pages 938-947. <https://doi.org/10.1016/j.applthermaleng.2016.12.058>
- [7] Miara M, Günther D, Langner R, Helmling S, Wapler J. 10 years of heat pump monitoring in Germany: Outcomes of several monitoring campaigns from low-energy houses to un-retrofitted single-family

- dwellings. In: Proceedings of the 12th IEA Heat Pump Conference; 2017.
<https://etkhhpcorderapi.extweb.sp.se/api/file/1642>
- [8] Heat Pump KEYMARK Certificates. Certified Products. Available from:
<https://keymark.eu/en/products/heatpumps/certified-products>. Accessed 12 February 2025.
- [9] Tjaden T, Hoops H, Rösken K. RE-Lab-Projects/hplib: Heat Pump Library (v2.0).
2021. <https://doi.org/10.5281/zenodo.5521597>. Accessed 12 February 2025.
- [10] Aste N, Adhikari RS, Manfren M. Cost optimal analysis of heat pump technology adoption in
residential reference buildings. *Renewable Energy*, Volume 60, 2013, Pages 615-624.
<https://doi.org/10.1016/j.renene.2013.06.013>
- [11] Bouzghiba H, Géczi G. Comparative analysis of the environmental load of natural refrigerant (R290).
Comparative Analysis of the Environmental Load of Natural Refrigerant (R290). *Journal of Central
European Green Innovation*, 10 (Suppl 1), 135-142, 2022. <https://doi.org/10.33038/jcegi.3506>